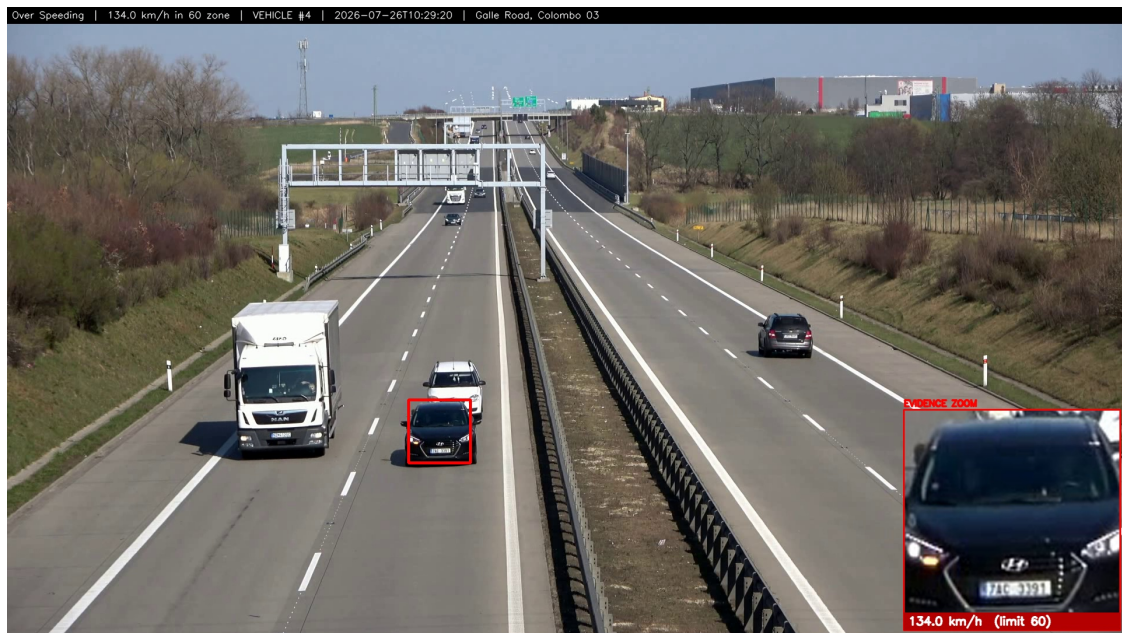


AI Smart Traffic Intelligence Platform

Technical Documentation and Verified Test Results



System output: court-ready over-speeding evidence, 134.0 km/h in a 60 km/h zone, with number-plate zoom inset.

Every figure in this document was measured on the development machine (12-core CPU, no GPU). Nothing is estimated.

1. What the system does

The platform turns an ordinary camera - CCTV, a phone stream or a drone - into an automated traffic officer. It detects and tracks every vehicle, judges nine categories of violation, reads number plates, measures real speed, and generates an e-challan carrying photographic evidence that is delivered to the police.

2. Detection pipeline

Camera feed → **YOLOv8s** detection → **ByteTrack** tracking with persistent per-vehicle IDs → **homography** mapping pixels to road metres → **violation engine** with false-positive defences → **EasyOCR** plate reading → evidence snapshot → e-challan.

3. The nine rules

Rule	How it is judged	Requires
No Helmet	Helmet model on each rider's own head region	Motorcycle + rider
Triple Riding	Riders associated with one motorcycle	3 riders detected
Over Speeding	Least-squares fit of road-metres against time	Camera calibration
Red Light Jump	Stop-line crossing while the governing signal is red	Signal ROI + lane zone
Wrong Way	Direction of travel against the policed lane	Direction + lane zone
No Seatbelt	Seatbelt model on the windscreen region	models/seatbelt.pt
Mobile Phone Use	Phone attributed to a rider or driver	Visible phone
Illegal Parking	Stationary beyond a threshold in a no-parking zone	Fixed camera
Driver Fatigue	Continuous driving without a qualifying break	Long observation

4. Measured results

Test	Recorded result
Automated test suite	79 tests passing across 7 suites
sample_1080p.mp4 (calibrated)	7 Over Speeding events, speeds 83-163 km/h
stream.mp4 (calibrated)	11 of 52 vehicles measured, max 106 km/h
IMG_6992.MOV (Sri Lanka)	150 vehicles; ANPR read AAG 4002 and BBJ 8752
srilanka.mp4 (parked bikes)	41 vehicles, 0 violations - a correct refusal
Single-image upload	No Helmet + Triple Riding, 2 challans issued
Real-time replay	0.99-1.01x true speed via frame dropping
Live webcam smoothness	1.89 to ~7 published fps after decoupling

5. Engineering for trustworthiness

A traffic system that issues wrong fines is worse than no system at all. Each defect below was found by testing against real footage, and each is now covered by the automated suite.

Defect found	Cause	Fix
Moving vehicles fined for parked	Motion gate needed 3 samples inside a 1-second frame-based interval at a 24px analysis; 89% of moving vehicles were	Time-based interval at a 24px analysis; 89% of moving vehicles were
Helmeted riders accused	Helmet model ran on a crop extending well beyond the bike-mounted rider's bounding region	Judge the bike-mounted rider's bounding region
Bare heads marked compliant	Claiming compliance needed only 0.35 confidence in the 50 longest signal	Decide by the 50 longest signal; raise the bar to 0.55
Stopped car fined for red light	Any light in frame was applied to any vehicle, including the car's own headlights	SUGAR-COOL picks the green signal; RED-LIGHT-ZONE limits
Wrong plate numbers on challenge	Two agreeing OCR reads allowed two wrong reads to agree by accident	Require three by accident
Triple riding never fired	Occluded pillion riders never cleared the 0.40 percent threshold	Lower threshold (0.25) for rider association only

6. Honesty by design

- Parked vehicles are never fined - the motion gate is a hard precondition.
- An unreadable plate reports **UNREADABLE**; a number is never invented.
- An uncalibrated camera shows **no speed** rather than a guess.
- A violation must persist across several frames; one-frame anomalies are rejected.
- There is no mock or seeded data anywhere in the product.

7. System output - verified evidence



Single-image analysis: three riders and no helmets detected on a CC-licensed public photograph, producing two separate challans.



Live Sri Lankan road: 18 vehicles tracked, ANPR reading plate AAG 4002, with per-vehicle speeds shown beside each ID.



Helmet judgement across mixed traffic.

8. Deployment

Runs on infrastructure that already exists: RTSP CCTV, a phone stream, or a drone via OBS. A pilot site needs one laptop; city density is served by GPU edge boxes. Helmet, seatbelt, plate and three-wheeler models are drop-in files requiring no code change.

Stack: Python, FastAPI, YOLOv8, ByteTrack, OpenCV, EasyOCR and SQLite. The dashboard is a single self-contained HTML file with no external dependencies.

9. Known limits

Stated plainly, because anyone deploying this must understand them:

- Speed requires per-camera calibration; without it no speed is shown.
- Plate OCR needs roughly 40 or more pixels of plate width.
- Red-light enforcement requires knowing which signal head governs the policed lane.
- On a CPU laptop about 2 frames per second are analysed; the video stays real-time by dropping frames, exactly as a live camera does.
- Detection degrades at night and in heavy rain, as with any camera system.

10. Roadmap

Night and adverse-weather models; a vision-language model such as Moondream to resolve the cases the detector currently reports as not judged; GPU edge deployment at a live junction; and a city-wide dashboard aggregating multiple cameras.